

ORIGINAL ARTICLE

Medications for Nausea and Vomiting of Pregnancy and the Risks for Adverse Birth Outcomes

Laure Sillis^{1,2,3}  | Essi Whaites Heinonen^{1,4} | Michael Ceulemans^{2,3}  | Veerle Foulon² | Yunjun Luo¹ | Diana Johnson¹ | Christina D. Chambers¹

¹Department of Pediatrics, University of California San Diego, La Jolla, California, USA | ²Department of Pharmaceutical and Pharmacological Sciences, KU Leuven, Leuven, Belgium | ³Research Foundation Flanders, Brussels, Belgium | ⁴Department of Clinical Science, Intervention and Technology, Karolinska Institute, Stockholm, Sweden

Correspondence: Laure Sillis (laure.sillis@kuleuven.be)

Received: 22 August 2025 | **Revised:** 18 January 2026 | **Accepted:** 23 March 2026

Keywords: birth outcomes | doxylamine | hyperemesis gravidarum | nausea and vomiting of pregnancy | observational research | ondansetron | pharmacoepidemiology

ABSTRACT

Purpose: Nausea and vomiting of pregnancy (NVP) is common during pregnancy. However, evidence on the safety of its treatment is still conflicting. This study compared the risk for adverse birth outcomes between mothers exposed to NVP medications and unmedicated mothers, with and without NVP.

Methods: This prospective, observational cohort study is part of the MotherToBaby Pregnancy Studies initiative, which recruits participants during pregnancy and collects data through maternal telephone interviews and medical records. The study includes live-born singleton pregnancies enrolled from 2010 to 2023, with available first-trimester NVP symptom data. Pregnant women exposed to any NVP medications were compared to pregnant women with untreated NVP and pregnant women without NVP. Statistical analysis involved calculating risk ratios (RR) for major congenital malformations, preterm delivery, small for gestational age (SGA), and admission to the neonatal intensive care unit (NICU).

Results: This study included 2711 pregnant individuals, categorized into NVP treatment exposed ($n = 603$; mean [SD] age, 33.2 [4.3] years), untreated NVP ($n = 1567$; mean [SD] age, 33.3 [4.4] years), and no NVP ($n = 541$; mean [SD] age, 33.4 [4.4] years). Major congenital malformations occurred in 6.12% ($n = 29$) of the NVP treatment exposed group, 5.11% ($n = 80$) of the untreated NVP group, and 5.55% ($n = 30$) of the no NVP group, with adjusted RR for treatment exposed 1.22 (95% CI, 0.79–1.87) compared to untreated NVP and 1.05 (95% CI, 0.63–1.76) compared to no NVP. The adjusted RR of preterm birth was 1.21 (95% CI, 0.88–1.67) for treatment exposed compared to untreated NVP. The adjusted RR of infants being SGA was 1.25 (95% CI, 0.90–1.74) compared to untreated NVP. Doxylamine ($n = 345$ [57.21%]) and ondansetron ($n = 286$ [47.43%]) were the most commonly used NVP treatments. When the risks for adverse birth outcomes were evaluated separately for these medications, ondansetron exposure was associated with higher RRs for SGA compared with both untreated NVP (adjusted RR 1.80, 95% CI 1.23–2.60) and no NVP (1.99, 95% CI 1.22–3.24). The confidence intervals indicate that the estimates are relatively imprecise but suggest an increased risk in both comparisons.

In this manuscript, we use the terms “women,” “woman,” “mother,” and “maternal” when referring to people who are pregnant or who give birth, in line with common clinical and epidemiological terminology. The authors acknowledge that pregnancy and childbirth occur among people with diverse gender identities, and not all individuals who experience pregnancy or birth identify as women.

Preliminary findings from this study were shared as a poster at the 13th Annual Pediatric Research Symposium 2024 (San Diego). Results were subsequently presented as a spotlight poster at the 41st Annual Meeting of the International Society for Pharmacoepidemiology (ISPE) 2025 (Washington, DC) and presented orally at the Joint ENTIS-OTIS Meeting 2024 (Nyborg).

Conclusions: No increased risks of major congenital malformations or other adverse birth outcomes were observed when comparing women treated with NVP medications to those with untreated NVP or no NVP, in overall comparisons, supporting the pharmacological treatment of NVP when needed. Ondansetron exposure was associated with an increased risk of SGA, which may at least partially be influenced by greater NVP severity among ondansetron-treated women rather than treatment effect itself.

1 | Introduction

Nausea and vomiting of pregnancy (NVP) affects around 70% of pregnant women with varying severity [1]. Severe cases of NVP can lead to hyperemesis gravidarum (HG), which occurs in 0.3%–3.6% of pregnancies [1, 2]. NVP symptoms mainly occur in the first trimester, beginning at approximately 6–8 weeks of gestation and generally subsiding by weeks 16–20 [3]. While often considered a benign and self-limiting condition, the presence and severity of NVP seriously impacts the mother's health-related quality of life [4]. Severe NVP or HG is the leading cause of hospitalization in the first trimester and the second most common reason for pregnancy hospitalization overall [5]. Additionally, HG is associated with poor maternal, fetal, and child outcomes [2].

The management of NVP involves pharmacological treatments aimed at alleviating the symptoms, improving the quality of life, and preventing progression to severe NVP or HG. However, the safety of NVP medications during pregnancy remains a critical concern, particularly regarding their potential impact on fetal development and birth outcomes. NVP symptoms mainly present in the first trimester, the most critical period for organogenesis. Current evidence on treatment safety and effectiveness is still conflicting, resulting in varying international guidelines [6, 7]. A 2010 Cochrane review assessing the effectiveness and safety of interventions for NVP concluded that data on maternal and fetal adverse outcomes after exposures to NVP treatments are insufficient [8], and newer systematic reviews or meta-analyses are lacking. Recent studies evaluating the safety of NVP treatments have presented with conflicting results, particularly regarding a potentially increased risk of major congenital malformations after in utero exposure to doxylamine [9–12] or ondansetron [13–15]. However, the studies suggesting an increased risk of major congenital malformations have been questioned due to confounding by indication, the inherent risks of using “big data,” such as detecting false signals with high confidence [16], and the limitations of case–control designs.

Previous studies evaluating the safety of NVP treatments have often not studied the impact of NVP itself. However, NVP and its severity could impact outcomes for both mother and infant, both positively and negatively. A population-based cohort study found that women with NVP were more likely to develop pregnancy complications, but did not show negative birth outcomes or complications for the infant [17]. Similarly, a recent case–control study reported no effect or a small protective effect on birth outcomes such as low birth weight and small for gestational age (SGA) for first trimester NVP and short-lasting NVP [18]. However, long-lasting and severe NVP may increase the risks of preterm birth and the infant being SGA [18]. Recently, a large case–control study found that NVP appeared to be largely protective against birth defects [12]. However, the use of prescription and over the counter (OTC) antiemetics was associated

with higher risks of specific birth defects, including tetralogy of Fallot, hypoplastic left heart syndrome, gastroschisis, and craniosynostosis. The authors used prescription treatments compared with OTC treatments as a proxy for NVP severity and concluded that the increases in risks did not seem to be higher for prescription medications compared to OTC treatments.

To further address these issues, we aimed to assess the risks of major congenital malformations, preterm birth, the infant being SGA, and admission to the neonatal intensive care unit (NICU) for infants born to women who were exposed to NVP medications compared to women with untreated NVP and women without NVP, using prospectively collected data in the MotherToBaby Pregnancy Studies.

2 | Methods

2.1 | Study Design and Population

This prospective, observational cohort study utilized data from the MotherToBaby Pregnancy Studies, conducted by the Organization of Teratology Information Specialists (OTIS) in the United States and Canada. All participants were recruited from these two countries. Pregnant women enrolled between July 2010 and December 2023 were included. Data were collected through maternal interviews and medical record extractions. Participant recruitment methods and data collection procedures are detailed in the [Supporting Information](#) and have been described previously [19]. Only pregnancies resulting in live-born singletons were included. Participants were included if information on NVP symptoms in the first trimester (from conception until the end of gestational Week 12) was available. This meant that, they had either completed the Pregnancy-Unique Quantification of Emesis and Nausea (PUQE) questionnaire [20] in the first trimester or answered general questions about their NVP symptoms (e.g., “Have you experienced any pregnancy-related nausea?”, “Have you experienced any pregnancy-related vomiting?”, “Have you been diagnosed with hyperemesis?”) during the first trimester. All participants provided informed consent, and the study received Institutional Review Board approval from the University of California, San Diego.

2.2 | Exposure Assessment

Pregnant women exposed to NVP medications were compared to those with untreated NVP and those without NVP. NVP was defined as a PUQE Score of 3 or more or a positive answer to one or more general NVP questions in the first trimester. Participants who answered both the PUQE and general NVP questions were classified as having NVP if they met the criteria in either. Participants exposed to NVP treatment without registered NVP symptoms were classified as exposed if the

Plain Language Summary

Nausea and vomiting are very common during pregnancy, affecting about 70% of pregnant individuals. Although medications are available to relieve these symptoms, there is still uncertainty about whether they are safe for the unborn child. This study, part of the MotherToBaby Pregnancy Studies initiative, followed over 2700 pregnant individuals between 2010 and 2023. We compared birth outcomes among those who took nausea and vomiting medications, those who had symptoms but didn't take medication, and those who didn't experience nausea. In overall analyses of all nausea and vomiting medications, we did not find an increased risk of birth defects, premature birth, or infants being small for their gestational age. The most commonly used treatments were doxylamine and ondansetron. When looking at pregnancies exposed to ondansetron, we found a higher chance of having an infant being small for gestational age. However, this may be due to more severe nausea rather than the medication itself. Overall, our findings provide reassuring evidence that nausea and vomiting treatments can be safely used during pregnancy when needed.

Key Points

- Nausea and vomiting of pregnancy (NVP) affects around 70% of pregnant individuals, yet treatment safety remains unclear.
- At the group level, when comparing individuals treated with any NVP medication to those with untreated NVP or no NVP, no increased risk for major malformations, preterm birth, or the infant being small for gestational age (SGA) was observed.
- Ondansetron exposure was associated with a higher risk of SGA; however, women treated with ondansetron had higher rates of vomiting and hyperemesis compared with the overall NVP-exposed group, suggesting that this association may at least partially be influenced by confounding due to greater NVP severity.
- This study offers reassuring evidence that pharmacological treatment can be safely used by pregnant women experiencing NVP without increasing the risk of adverse birth outcomes.

treatment indication was documented as NVP. Those without NVP symptoms but exposed to treatments for other indications were excluded.

All medications indicated or recommended for NVP by the United States medication database Clinical Pharmacology (ClinicalKey/Elsevier) were included as NVP treatments [21]. These included diphenhydramine, dimenhydrinate/dramamine, doxylamine (succinate), ginger (*Zingiber officinale*), metoclopramide, ondansetron, phosphorated carbohydrate solution, promethazine, prochlorperazine, and pyridoxine (vitamin B6). Participants were considered exposed if they received NVP treatment in the first trimester for major malformations

analysis, and anytime during pregnancy for other outcomes. The Slone Epidemiology Center's Drug Dictionary was used to code and identify exposures from the database. Data on exposures, including indication, timing, and dose, were collected during semi-structured maternal interviews. Exposures to Benadryl (diphenhydramine, marketed to relieve hay fever and other upper respiratory tract allergies) and methylprednisolone were not considered NVP treatments due to their typical indications. To minimize misclassification, pregnancies exposed to NVP treatment both before and during pregnancy and those with missing exposure timing data were excluded.

2.3 | Outcome Measures

The main outcome measures were major congenital malformations, preterm delivery, small for gestational age at birth (SGA), and admission to the NICU. Outcome data were collected through postpartum maternal interviews and verified by medical records from obstetricians, pediatricians, delivery hospitals, and relevant pathology reports when available. Major congenital malformations were defined as those with cosmetic or functional significance to the child, identified up to 1 year after birth by the mother, healthcare provider, medical record, or dysmorphological examination. Malformations were classified according to the CDC's Metropolitan Atlanta Congenital Defects Program coding manual [22] and reviewed by a dysmorphologist or geneticist. Preterm delivery was defined as delivery before 37 weeks of gestation, with ultrasound dating used to correct gestational age when necessary. SGA was defined as a birth weight $\leq$ 10th percentile for sex and gestational age on US growth charts [23]. NICU admission was used as a proxy for general neonatal morbidity, reported by either the mother or medical record.

Covariates included maternal age, race and ethnicity, Hollingshead Category of Socioeconomic Status (SES) [24], pre-pregnancy BMI, obstetric history, prenatal multivitamin or folic acid supplement use, exposure to known human teratogens during pregnancy, and substance use. These covariates were collected through maternal telephone interviews and are detailed in the [Supporting Information](#).

2.4 | Statistical Analysis

A comparison was conducted between women exposed to NVP medications and a disease-matched group with untreated NVP, followed by a comparison to women without NVP. For the analysis of major congenital malformations, the exposed group was restricted to those treated in the first trimester (Figure 1). Participant characteristics were presented as means and standard deviations (SD) or numbers and percentages. To describe differences in participant characteristics between exposure groups, standardized differences were calculated (see [Supporting Information](#) for detailed methods). Risk ratios (RR) and 95% confidence intervals (CIs) for selected outcomes were calculated using modified Poisson regression with robust error variance. Robust variance estimators were applied because the Poisson model was used for binary outcomes, and robust variance corrects the standard errors for this model misspecification. To adjust for confounding, a propensity score was computed

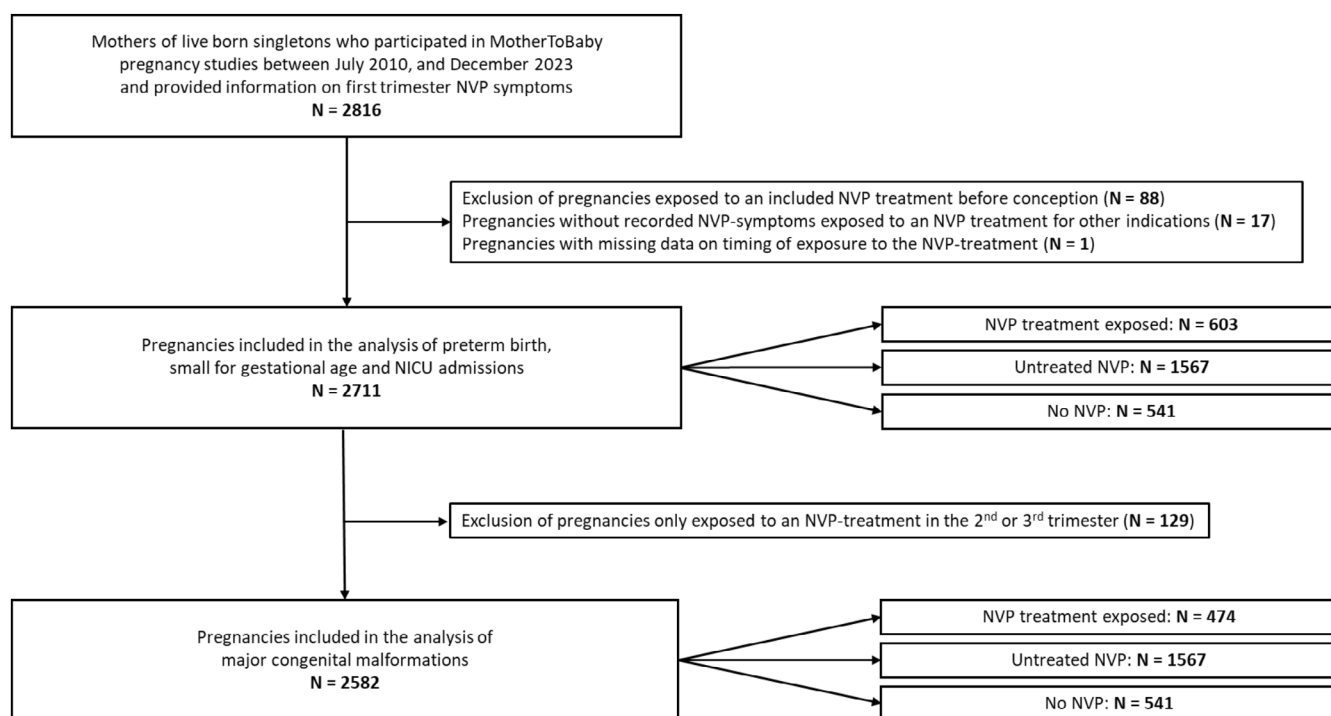


FIGURE 1 | Flow diagram of study participants. *Abbreviations:* NICU: neonatal intensive care unit; NVP: nausea and vomiting of pregnancy.

using logistic regression including the following covariates entered as main effects: maternal age (categorized as <25, 25–29, 30–34, >34 years), Hollingshead socioeconomic status (high vs. low), gravidity (1 vs. >1), pre-pregnancy BMI (<18.5, 18.5–24.9, 25–29.9, >30 kg/m²), race/ethnicity (White non-Hispanic, Hispanic/Latina, Black non-Hispanic, Other), and self-reported exposures to smoking, alcohol, and recreational substances during pregnancy (each coded as yes/no) (Table 1). The resulting propensity score was included in the modified Poisson regression models as a continuous variable, together with the binary exposure variable. Participants with missing data on any confounder were excluded from the adjusted analysis. Missing data are presented in Table 1. In a sensitivity analysis, covariates were adjusted individually rather than using a propensity score. A second sensitivity analysis examined the effect of disease severity, differentiating between NVP monotherapy (use of one NVP treatment) and polytherapy (use of more than one NVP treatment) throughout pregnancy. In this analysis, outcomes were compared among women exposed to NVP monotherapy and polytherapy to untreated NVP, and no NVP. A third sensitivity analysis evaluated the risk for preterm birth, SGA, and NICU admission restricting the exposed group to those treated in the first trimester. All analyses were performed with IBM SPSS Statistics 29.0.2.0, and GraphPad Prism 10.3.1. was used for data visualizations.

3 | Results

Of the 2816 women participating in the MotherToBaby registry during the study period who gave birth to a live-born singleton infant and provided information on first-trimester NVP, 106 (3.8%) were excluded (Figure 1). The remaining 2711 women were categorized into NVP treatment exposed ($n=603$; mean

[SD] age, 33.2 [4.3] years), untreated with NVP ($n=1567$; mean [SD] age, 33.3 [4.4] years), and no NVP ($n=541$; mean [SD] age, 33.4 [4.4] years). Among women treated for NVP, treatment was initiated at a mean (SD) gestational age of 11.1 (7.9) weeks. Of the 603 women in the NVP treatment exposed group, 129 (21.4%) were exposed only during the second or third trimester. For the analysis of major malformations, the exposed group was limited to the 474 pregnancies exposed in the first trimester (Figure 1).

The background characteristics of the included women are presented in Table 1. The exposure groups were similar in terms of maternal age, ethnicity, Hollingshead SES Category, rates of having a previous child with a major congenital malformation, previous preterm birth, and pre-pregnancy BMI. The NVP treatment exposed group had a slightly higher proportion of women identifying as White, while both comparison groups had more women identifying as Black or Asian (Table 1). The group without NVP also had a higher proportion of primigravidae than both groups with NVP. Alcohol use during pregnancy was comparable across groups, and tobacco and recreational substance use were generally low with minimal differences between groups. Mean gestational age at enrollment was 16 weeks among individuals with treated or untreated NVP and 14 weeks among those without NVP.

The rates of major congenital malformations were 6.12% ($n=29$) in the NVP treatment exposed group, 5.11% ($n=79$) in the untreated NVP group and 5.55% ($n=30$) in the no NVP group (Table 2, and Table S1 for a list of all the specific major congenital malformations). The adjusted risks of major congenital malformations for the NVP treatment exposed group were 1.22 (95% CI, 0.79–1.87) compared to untreated NVP and 1.05 (95% CI, 0.63–1.76) compared to no NVP. In a sensitivity analysis, we excluded nine pregnancies affected by chromosomal or genetic

TABLE 1 | Characteristics of participants in each study group.

	NVP- treatment exposed (N= 603)	Untreated NVP (N= 1567)	Std. diff. versus untreated^a	No NVP (N= 541)	Std. diff. versus no NVP^a
<i>Age at estimated due date</i>					
< 25 years	19 (3.2%)	51 (3.3%)	0.01	15 (2.8%)	0.08
25–29 years	108 (17.9%)	272 (17.4%)	0.01	98 (18.1%)	0.01
30–34 years	213 (35.3%)	556 (35.5%)	0.00	206 (38.1%)	0.06
> 34 years	263 (43.6%)	688 (43.9%)	0.01	222 (41.0%)	0.06
<i>Race</i>					
Asian	21 (3.5%)	91 (5.8%)	0.11	42 (7.8%)	0.26
Black	17 (2.8%)	68 (4.3%)	0.09	29 (5.4%)	0.09
Indian/Native American	14 (2.3%)	16 (1.0%)	0.10	7 (1.3%)	0.07
Pacific Islander	1 (0.2%)	3 (0.2%)	0.01	0 (0.0%)	0.07
White	534 (88.6%)	1318 (84.2%)	0.13	428 (79.1%)	0.26
Other	11 (1.8%)	35 (2.2%)	0.03	18 (3.3%)	0.10
Missing	5 (0.8%)	36 (2.3%)		17 (3.1%)	
<i>Ethnicity</i>			0.03		0.07
Non-Hispanic/Non-Latina	550 (91.2%)	1435 (91.6%)		481 (88.9%)	
Hispanic/Latina	53 (8.8%)	127 (8.1%)		58 (10.7%)	
Missing	0 (0.0%)	5 (0.3%)		2 (0.4%)	
<i>Hollingshead SES category^b</i>			0.01		0.08
Low	25 (4.1%)	69 (4.4%)		32 (5.9%)	
High	571 (94.7%)	1485 (94.8%)		506 (93.5%)	
Missing	7 (1.2%)	13 (0.8%)		3 (0.6%)	
<i>Pre-pregnancy BMI category^c</i>					
< 18.5	16 (2.7%)	41 (2.6%)	0.00	14 (2.6%)	0.00
18.5–24.9	310 (51.8%)	807 (51.9%)	0.00	313 (58.4%)	0.13
25–29.9	162 (27.0%)	370 (23.8%)	0.07	117 (21.8%)	0.12
≥ 30	111 (18.5%)	337 (21.7%)	0.08	92 (17.2%)	0.04
Missing	4 (0.7%)	12 (0.8%)		5 (0.9%)	
<i>Gravidity category</i>			0.02		0.18
1	226 (37.5%)	573 (36.5%)		250 (46.2%)	
> 1	377 (62.5%)	994 (63.4%)		291 (53.8%)	
<i>Previous preterm delivery</i>	28 (4.6%)	97 (6.2%)	0.06	36 (6.7%)	0.09
<i>Any previous child with a major congenital malformation</i>	36 (6.0%)	60 (3.8%)	0.09	25 (4.6%)	0.06
<i>Prenatal multivitamin or folic acid</i>			0.02		0.01
Prior to conception	441 (73.1%)	1125 (71.7%)		397 (73.4%)	

(Continues)

TABLE 1 | (Continued)

	NVP-treatment exposed (N = 603)	Untreated NVP (N = 1567)	Std. diff. versus untreated ^a	No NVP (N = 541)	Std. diff. versus no NVP ^a
Post-conception only, or not taken	162 (26.9%)	440 (28.1%)		144 (26.6%)	
Missing	0 (0.0%)	2 (0.1%)		0 (0.0%)	
<i>Substance use during pregnancy</i>					
Alcohol use	267 (44.3%)	699 (44.6%)	0.02	245 (45.3%)	0.02
Tobacco use	11 (1.8%)	54 (3.4%)	0.11	19 (3.5%)	0.11
Recreational drug	30 (5.0%)	58 (3.7%)	0.06	10 (1.9%)	0.17
Missing recreational drug use	0 (0.0%)	0 (0.0%)		1 (0.2%)	
Exposure to known human teratogens during pregnancy ^d	6 (1.0%)	5 (0.3%)	0.16	2 (0.4%)	0.08
Gestational age at enrollment (weeks), mean (SD)	16.2 (8.6)	16.3 (8.7)	0.01	13.9 (7.9)	0.28

Abbreviations: NVP, nausea and vomiting of pregnancy; SD, standard deviation; SES, socio economic status; Std diff: standardized difference (absolute).

^aFor estimation of the absolute standardized difference equation, see 2.4 Statistical Analysis section.

^bHollingshead SES categories are defined as categories 1–5, in which high SES = categories 1, 2, or 3 and low SES = category 4 or 5.

^cThe pre-pregnancy BMI categories were defined based on specific ranges of BMI (in kg/m²).

^dExposure to known human teratogens in pregnancy here includes exposure to valproic acid, topiramate, coumadin, warfarin, lithium, methimazole, methotrexate, cyclophosphamide, cocaine, chemotherapy, heavy drinking (≥ 14 drinks per week for > 4 weeks) in pregnancy, fever (102°F/37.8°C or higher for 24 h or longer) in pregnancy, or a diagnosis of Type 1 or 2 diabetes.

major congenital malformations (2 in the NVP treated group, 5 in the untreated NVP group, and 2 in the no NVP group). The adjusted risk ratios for major congenital malformations were essentially unchanged: 1.21 (95% CI, 0.77–1.88) when comparing NVP-treated pregnancies with untreated NVP, and 1.02 (95% CI, 0.60–1.75) when comparing NVP-treated pregnancies with no NVP.

The rates of preterm birth were 8.62% ($n = 52$) in the exposed group, compared to 7.22% ($n = 113$) in the untreated NVP group and 7.95% ($n = 43$) in the no NVP group. For preterm birth, adjusted RRs of 1.21 (95% CI, 0.88–1.67) and 1.09 (95% CI, 0.74–1.61) were found, compared to untreated NVP and no NVP, respectively. The rates of infants being SGA were similar between groups, with adjusted RRs of 1.25 (95% CI, 0.90–1.74) for exposed women compared to unexposed NVP and 1.42 (95% CI, 0.92–2.18) compared to no NVP. The risk of the infant being admitted to NICU was not increased for the exposed group (Table 2).

In a sensitivity analysis, adjusted RRs were calculated using modified Poisson regression, directly adjusting for selected covariates. The results were consistent with the primary analyses. Additionally, a sensitivity analysis was conducted to explore the effect of disease severity. However, the risks were similar in both women treated with one ($n = 461$ [76.45%]) and more than one ($n = 142$ [23.55%]) NVP medication during pregnancy, but the group sizes were small (Table S2).

As the majority of the NVP exposures were exposures to doxylamine ($n = 345$ [57.21%]) and ondansetron ($n = 286$ [47.43%]) (Table 3), the analyses were replicated specifically studying any

risks of these exposures in mono- or polytherapy. The results of these analyses are presented in Tables 4 and 5. The rate of major malformations was slightly higher in the doxylamine-exposed group than in the untreated NVP and no NVP groups ($n = 20$ [6.97%] compared to $n = 80$ [5.11%] and $n = 30$ [5.55%], respectively), but the adjusted RRs were close to the null with wide CI (1.32 [95% CI, 0.80–2.17] compared to untreated NVP and 1.07 [95% CI, 0.60–1.89] compared to no NVP). No other increased risks for doxylamine use were identified (Table 4). For ondansetron, no differences were found in the rates of major malformations, preterm birth or NICU admissions (Table 5). However, the rate of infants being SGA was higher in the ondansetron-exposed group ($n = 33$ [11.70%]), than in the untreated NVP ($n = 109$ [7.05%]) and no NVP groups ($n = 39$ [7.24%]), with adjusted RRs of 1.80 (95% CI, 1.23–2.60) compared to untreated NVP and 1.99 (95% CI, 1.22–3.24) to no NVP. Women exposed to ondansetron had the highest rates of both vomiting and HG compared to other exposure groups (Table 6).

4 | Discussion

4.1 | Principal Findings

In this prospective cohort study, exposure to NVP treatment during pregnancy was not associated with an increased risk of major malformations compared to groups with untreated NVP and those without NVP. Additionally, the risks for preterm birth, SGA infants, and NICU admissions were similar among infants born to women with treated NVP, untreated NVP, and no NVP. The similarity between crude and adjusted

TABLE 2 | Birth outcomes among women exposed to any NVP treatment (considering exposures in mono- or polytherapy) compared to women with untreated NVP and women without NVP.

	Counts and percentages in the study groups			Exposed versus untreated NVP		Exposed versus no NVP		
	NVP treatment exposed n/N (%)	Untreated NVP n/N (%)	No NVP n/N (%)	Unadjusted RR (95% CI)	Adjusted ^a RR (95% CI)	Unadjusted RR (95% CI)	Adjusted ^a RR (95% CI)	
Major malformations	29/474 ^b (6.12%)	80/1567 (5.11%)	30/541 (5.55%)	1.20 (0.80–1.81)	1.22 (0.79–1.87)	1.10 (0.67–1.81)	1.05 (0.63–1.76)	● Compared to untreated NVP ○ Compared to no NVP
Preterm birth	52/603 (8.62%)	113/1566 (7.22%)	43/541 (7.95%)	1.200 (0.87–1.64)	1.21 (0.88–1.67)	1.09 (0.74–1.60)	1.09 (0.74–1.61)	
SGA	52/593 (8.77%)	109/1546 (7.05%)	39/539 (7.24%)	1.24 (0.91–1.71)	1.25 (0.90–1.74)	1.21 (0.81–1.81)	1.42 (0.92–2.18)	
NICU admission	66/599 (11.02%)	162/1552 (10.44%)	59/537 (10.99%)	1.06 (0.81–1.39)	1.09 (0.83–1.43)	1.00 (0.72–1.40)	1.06 (0.74–1.50)	

Abbreviations: CI, confidence interval; NICU, neonatal intensive care unit; NVP, nausea and vomiting of pregnancy; preterm birth, birth at < 37.0 weeks of gestation; RR, risk ratio; SGA, small for gestational age (birth weight ≤ 10th percentile).
^aAdjusted for maternal age, Hollingshead socioeconomic status category, race, ethnicity, gravidity, pre-pregnancy body mass index, and exposure to alcohol, smoking and recreational substances in pregnancy.
^bOnly pregnancies exposed to an NVP treatment in the 1st trimester were included in the NVP treatment exposed group for the analysis of major malformations.

TABLE 3 | Counts and percentages of exposures to specific NVP treatments in the total pregnancy and in the first trimester of pregnancy.

	Total pregnancy <i>n</i> (%)	First trimester <i>n</i> (%)
Doxylamine (succinate)	345 (57.21%)	287 (60.55%)
Ondansetron	286 (47.43%)	234 (49.37%)
Metoclopramide	46 (7.63%)	41 (8.65%)
Promethazine	42 (6.97%)	35 (7.38%)
Dimenhydrinate/Dramamine	37 (6.14%)	24 (5.06%)
Ginger, <i>Zingiber officinale</i>	10 (1.66%)	9 (1.90%)
Prochlorperazine	5 (0.83%)	3 (0.63%)
Phosphorated carbohydrate solution	2 (0.33%)	1 (0.21%)

RRs across all birth outcome analyses suggests that the associations were not influenced by the included confounding variables. In contrast to the overall findings for NVP treatment, ondansetron exposure was associated with a higher risk of SGA.

4.2 | Results

These findings differ from a case-control study by the US National Birth Defects Prevention Study, which reported higher risks of specific congenital malformations in offspring of mothers who used NVP treatments compared to those with untreated NVP [12]. The prevalence of NVP treatment-use in the case-control study was significantly lower than in our cohort. Potential recall and selection biases, the known limitations of the case-control design, may explain these differing results. Additionally, our study did not have the statistical power to examine specific congenital defects. While the impact of NVP on birth outcomes has been studied elsewhere [17, 18, 25], the specific risks of preterm birth, the infant being SGA, and NICU admissions when directly comparing pregnancies with treated versus untreated NVP, as presented in this study, had not been previously examined.

Analyses of women exposed to doxylamine (in mono- or polytherapy), in our study, showed no increased risks of the studied outcomes, consistent with earlier findings [26]. For women exposed to ondansetron (in mono- or polytherapy), no increased risk for major congenital malformations was found compared to untreated NVP and no NVP, aligning with previous research [27]. However, due to the limited sample size, our study was underpowered to clarify the risk for specific major congenital malformations which each have low absolute risks, including malformations such as oral clefts that have previously been associated with ondansetron exposure [10, 14, 28].

4.3 | Clinical Implications

In this study, a positive association was found between exposure to ondansetron, in both mono- and polytherapy, and the infants being SGA. Although one previous study using electronic health data also found a weak association between ondansetron

exposure and SGA infants (HR=1.37, 95% CI 0.98–1.90), subsequent analyses of mean birth weights showed no differences [29]. A Danish population-based study reported no association between ondansetron use and low birth weight (<2500g) after adjusting for hospitalizations for HG or NVP [30]. In contrast to the MotherToBaby sample, ondansetron exposure was rare in the Danish cohort (0.3%), and over 50% of those exposed were hospitalized for HG or NVP [30]. It is known that severe NVP and especially HG may increase the risk for decreased fetal growth [31–33], possibly due to undernutrition of the pregnant woman. In our sample, the women exposed to ondansetron had higher rates of both vomiting and HG compared to the overall exposed group (Table 6), suggesting that uncontrolled confounding by disease severity, rather than the medication itself, could explain the increased risk for their infants being born SGA. Although fetal growth is often associated with exposures later in pregnancy, the elevated risk of SGA associated with ondansetron persisted after restricting the analysis to pregnancies with first-trimester ondansetron exposure (thereby excluding women treated only in the second or third trimester), suggesting that this association is unlikely to be explained solely by late-pregnancy exposure. This finding may reflect residual confounding related to greater NVP severity among ondansetron-treated women. Sample size restricted further exposure window stratification in this study. We cannot exclude the possibility that this result is due to multiple testing, or other unmeasured confounding, and highlight the need for future studies with larger sample sizes to more comprehensively investigate exposure window specific effects on SGA.

Pregnant women often hesitate to use NVP treatments [34]. A Norwegian survey (*n*=712) found that 70.2% of women with moderate NVP and 32.9% with severe NVP did not receive treatment, even when on sick leave [35]. Our study offers reassuring safety evidence to guide women and healthcare professionals in making informed decisions about NVP treatment.

4.4 | Research Implications

By its design, the study had the potential to adjust for NVP severity using the PUQE scale. Unfortunately, the PUQE questions were answered by less than half of the total cohort (Table 6), making further severity adjustments impossible.

TABLE 4 | Birth outcomes among women exposed to doxylamine (considering exposures in mono- or polytherapy) in the first trimester compared to women with untreated-NVP and women without NVP.

	Counts and percentages in the study groups				Doxylamine exposed versus untreated		Doxylamine exposed versus no-NVP		● Compared to untreated NVP ○ Compared to no NVP
	Doxylamine exposed n/N (%)	Untreated NVP n/N (%)	No NVP n/N (%)		Unadjusted RR (95% CI)	Adjusted ^a RR (95% CI)	Unadjusted RR (95% CI)	Adjusted ^a RR (95% CI)	
Major malformations	20/287 ^b (6.97%)	80/1567 (5.11%)	30/541 (5.55%)		1.37 (0.85–2.19)	1.32 (0.80–2.17)	1.26 (0.73–2.17)	1.07 (0.60–1.89)	
Preterm birth	25/345 (7.25%)	113/1566 (7.22%)	43/541 (7.95%)		1.00 (0.66–1.52)	1.05 (0.67–1.61)	0.91 (0.57–1.47)	0.94 (0.58–1.53)	
SGA	23/338 (6.80%)	109/1546 (7.05%)	39/539 (7.24%)		0.97 (0.62–1.49)	0.96 (0.61–1.51)	0.94 (0.57–1.55)	1.17 (0.68–2.01)	
NICU admission	59/337 (10.99%)	162/1552 (10.44%)	59/537 (10.99%)		0.92 (0.65–1.32)	0.96 (0.67–1.39)	0.87 (0.58–1.31)	0.88 (0.57–1.36)	

Abbreviations: CI, confidence interval; NICU, neonatal intensive care unit; NVP, nausea and vomiting of pregnancy; preterm birth, birth at <37.0 weeks of gestation; RR, risk ratio; SGA, small for gestational age (birth weight ≤ 10th percentile).

^aAdjusted for maternal age, Hollingshead socioeconomic status category, race, ethnicity, gravidity, pre-pregnancy body mass index, and exposure to alcohol, smoking and recreational substances in pregnancy.

^bOnly pregnancies exposed to doxylamine in the 1st trimester were included in the doxylamine exposed group for the analysis of major malformations.

TABLE 5 | Birth outcomes among women exposed to ondansetron (considering exposures in mono- or polytherapy) in the first trimester compared to women with untreated-NVP and women without NVP.

	Counts and percentages in the study groups				Ondansetron exposed versus untreated		Ondansetron exposed versus no-NVP		
	Ondansetron exposed n/N (%)	Untreated NVP n/N (%)	No NVP n/N (%)		Unadjusted RR (95% CI)	Adjusted ^a RR (95% CI)	Unadjusted RR (95% CI)	Adjusted ^a RR (95% CI)	
Major malformations	12/234 ^b (5.13%)	80/1567 (5.11%)	30/541 (5.55%)		1.00 (0.56–1.81)	1.11 (0.61–2.03)	0.93 (0.48–1.77)	0.92 (0.47–1.80)	● Compared to untreated NVP ○ Compared to no NVP
Preterm birth	22/286 (7.69%)	113/1566 (7.22%)	43/541 (7.95%)		1.07 (0.69–1.65)	1.06 (0.68–1.66)	0.97 (0.59–1.59)	1.00 (0.61–1.65)	
SGA	33/282 (11.70%)	109/1546 (7.05%)	39/539 (7.24%)		1.66 (1.15–2.40)	1.80 (1.23–2.60)	1.62 (1.04–2.51)	1.99 (1.22–3.24)	
NICU admission	29/285 (10.18%)	162/1552 (10.44%)	59/537 (10.99%)		0.98 (0.67–1.43)	1.00 (0.68–1.46)	0.93 (0.61–1.41)	1.01 (0.64–1.57)	

Abbreviations: CI, confidence interval; NICU, neonatal intensive care unit; NVP, nausea and vomiting of pregnancy; preterm birth, birth at <37.0 weeks of gestation; RR, risk ratio; SGA, small for gestational age (birth weight ≤ 10th percentile).

^aAdjusted for maternal age, Hollingshead socioeconomic status category, race, ethnicity, gravidity, pre-pregnancy body mass index, and exposure to alcohol, smoking and recreational substances in pregnancy.

^bOnly pregnancies exposed to ondansetron in the 1st trimester were included in the ondansetron exposed group for the analysis of major malformations.

Adjusted RR

TABLE 6 | Counts and percentages of NVP symptoms in the first trimester.

	Total cohort (N = 2711) n (%)	NVP treatment exposed (N = 603) n (%)	Doxylamine exposed (N = 345) n (%)	Ondansetron exposed (N = 286) n (%)	Untreated NVP (N = 1567) n (%)
NVP symptoms in the 1st trimester	2170 out of 2711 (80.04%)				
PUQE questions completed	1237 out of 2711 (45.63%)	237 out of 603 (39.30%)	91 out of 345 (26.38%)	140 out of 286 (48.95%)	648 out of 1567 (41.35%)
No NVP (≤ 3)	387 (31.29%)	31 (13.08%)	4 (4.40%)	25 (17.86%)	4 (0.62%)
Mild (≤ 6)	554 (44.79%)	101 (42.62%)	43 (47.25%)	55 (39.29%)	453 (69.91%)
Moderate (7–12)	286 (23.12%)	101 (42.62%)	43 (47.25%)	58 (41.43%)	185 (28.55%)
Severe (≥ 13)	10 (0.81%)	4 (1.69%)	1 (1.10%)	2 (1.43%)	6 (0.93%)
General NVP questions completed	1500 out of 2711 (55.33%)	373 out of 603 (61.86%)	259 out of 345 (133.20%)	148 out of 286 (51.75%)	936 out of 1567 (59.73%)
Experienced nausea	1291 (86.88%)	363 (97.84%)	252 (98.05%)	145 (97.97%)	928 (99.36%)
Missing	14 (0.94%)	2 (0.54%)	2 (0.77%)	0 (0.00%)	2 (0.21%)
Experienced vomiting	646 (43.07%)	250 (68.12%)	163 (64.17%)	122 (82.99%)	396 (43.14%)
Missing	24 (1.60%)	6 (1.61%)	5 (1.93%)	1 (0.68%)	18 (1.92%)
Information on HG available	1974 out of 2711 (72.8%)	466 out of 603 (77.3%)	304 out of 345 (88.1%)	190 out of 286 (66.43%)	1180 out of 1567 (75.3%)
Experienced HG	29 (1.47%)	20 (4.29%)	6 (1.97%)	19 (10.0%)	9 (0.76%)

Abbreviations: HG, hyperemesis gravidarum; NVP, nausea and vomiting of pregnancy; PUQE, pregnancy-unique quantification of emesis and nausea (PUQE) questionnaire.

Nonetheless, a sensitivity analysis, which categorized the exposed group into mono- and polytherapy groups as a proxy for disease severity, produced results consistent with the overall analysis (Table S2). Future research should prioritize addressing NVP severity when evaluating the safety of NVP treatments, also to validate the SGA findings associated with ondansetron exposure. Additionally, the study's limited number of specific NVP treatment exposures restricted treatment-specific analyses to the two most common exposures (doxylamine and ondansetron); this scope could be expanded in future research.

4.5 | Strengths and Limitations

One of the strengths of this study was the collection of detailed information on medication use during pregnancy through multiple telephone interviews, providing in-depth data on both medication exposures and NVP symptoms. Additionally, the availability of comprehensive data on NVP symptoms enabled comparisons with a disease-matched cohort, which is often missing in studies on NVP medication safety.

Our study was limited by many women not reporting the indication for antiemetic treatment, relying instead on NVP questionnaires for indication confirmation. This could lead to misclassification if antiemetics were used for other reasons (e.g., allergies). As approximately half of participants reported NVP symptoms using the PUQE questionnaire in the first trimester, which captures symptoms only during the preceding 12h, while the remainder answered general questions on first-trimester NVP, the exact gestational timing of NVP onset could not be consistently determined. For women who completed the PUQE assessment, misclassification cannot be ruled out, as PUQE may not fully reflect NVP if the assessment occurred prior to onset or during periods when symptoms were not present in the 12h preceding the interview. Furthermore, incomplete data on early pregnancy terminations (potentially due to mal-developing fetuses or HG) limited the cohort to live-born infants, potentially underestimating risks associated with major congenital malformations, as only the prevalence in live births was assessed. The reliance on a volunteer sample may limit generalizability, a common issue in studies requiring self-reported data. Nonetheless, prospective recruitment before pregnancy outcomes were known minimized volunteer bias. Because most NVP medications in this cohort were initiated early in pregnancy (Table 3), the amount of unexposed time before treatment was limited, reducing—but not entirely eliminating—the potential for immortal time bias for birth outcomes. Moreover, a sensitivity analysis with restriction to first-trimester initiation (Table S3) attenuated the associations toward the null, suggesting that pregnancies initiating treatment later in pregnancy represent a higher-risk subgroup and that time-dependent confounding by NVP severity rather than immortal time bias might be a source of uncontrolled bias. Comprehensive data collection on covariates helped address potential confounders, supporting internal validity, although residual confounding due to unknown or unmeasured variables cannot be excluded, especially given the still incomplete understanding of the etiology of NVP and HG. Finally, the sample sizes of prospective pregnancy registries, including

the MotherToBaby cohort, are often insufficient to rule out elevated risks for specific major birth defects, highlighting the need for cooperation between registries.

5 | Conclusions

In this study using data from the prospective observational MotherToBaby cohort, no increased risks of major congenital malformations, preterm birth, infant being SGA, or NICU admissions were found for women exposed to any NVP medications during pregnancy compared to those with untreated NVP and those without NVP. In this cohort, the most commonly used NVP treatments were doxylamine and ondansetron. When the risks for adverse birth outcomes were evaluated separately for ondansetron-exposed pregnancies, an increased risk of SGA was observed, potentially at least partly linked to the severity of NVP rather than the treatment itself, as higher HG and vomiting rates were observed among those treated with ondansetron. This evidence on the safety of the most commonly used NVP medications in this cohort is reassuring and important for both individuals experiencing NVP and requiring pharmacological treatment as well as their caregivers. However, more data are needed to explore the determinants behind the increased frequency of SGA found in infants to mothers treated with ondansetron.

Author Contributions

Laure Sillis: conceptualization, methodology, validation, formal analysis, data curation, writing – original draft, visualization, project administration, funding acquisition. **Essi Whaites Heinonen:** conceptualization, methodology, validation, formal analysis, writing – review and editing, visualization. **Michael Ceulemans:** conceptualization, methodology, writing – review and editing. **Veerle Foulon:** conceptualization, methodology, writing – review and editing. **Yunjun Luo:** software, investigation, data curation, writing – review and editing, project administration. **Diana Johnson:** software, investigation, data curation, writing – review and editing, project administration. **Christina D. Chambers:** conceptualization, methodology, software, investigation, resources, writing – review and editing, visualization, supervision, project administration, funding acquisition.

Acknowledgments

We would like to thank all families who are participating in the MotherToBaby pregnancy studies and all staff at the Center for Better Beginnings, UC San Diego who make this research possible.

Funding

Laure Sillis and Michael Ceulemans are supported by the Research Foundation Flanders—FWO (1S35823N; 1246425N). Laure Sillis conducted this research during a stay at UC San Diego's Center for Better Beginnings, funded by a Research Foundation Flanders—FWO travel grant (V414124N). Essi Whaites Heinonen was funded by the Swedish Research Council (2023–0023) and the Söderström Königska Foundation (SLS-994869).

Ethics Statement

This study received approval from the Institutional Review Board (IRB) at the University of California San Diego (Study Number: 141162, last

renewal of IRB approval received April 16, 2024). All study participants provided informed consent.

Conflicts of Interest

No author received funding relevant to the present project. The disclosures below refer to funding unrelated to the current study. Christina D. Chambers receives research funding from the following industry sponsors and a foundation: Amgen, AstraZeneca, GlaxoSmithKline, Janssen Pharmaceuticals, Pfizer Inc., Regeneron, Hoffman La-Roche-Genentech, Genzyme Sanofi-Aventis, Takeda Pharmaceutical Company Limited, Sanofi, UCB Pharma, USA, Leo Pharma, Sun Pharma Global FZE, Gilead, Novartis, and the Gerber Foundation. Michael Ceulemans and Veerle Foulon coordinate the BELpREG pregnancy registry in Belgium, which has received independent research grants from P&G Health, UCB, Tilman, Almirall, Ceres Pharma, Sanofi, Effik, and Johnson & Johnson. The KU Leuven/BELpREG team conducted an Effik-sponsored drug utilization study on doxylamine use during pregnancy using BELpREG data; funding was provided as research support to BELpREG, no author received personal remuneration, and the study was independent of the present work.

Data Availability Statement

The data used for this data analysis were originally collected under participants' informed consent for uses as directed by the study principal investigator and are not available for public access. Requests for use of the data should be directed to Christina Chambers, the senior researcher, at chchambers@health.ucsd.edu. This manuscript is aligned with guidance from the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) checklist.

References

1. T. R. Einarson, C. Piwko, and G. Koren, "Quantifying the Global Rates of Nausea and Vomiting of Pregnancy: A Meta Analysis," *Journal of Population Therapeutics and Clinical Pharmacology* 20, no. 2 (2013): e171–e183.
2. M. S. Fejzo, J. Trovik, I. J. Grooten, et al., "Nausea and Vomiting of Pregnancy and Hyperemesis Gravidarum," *Nature Reviews Disease Primers* 5, no. 1 (2019): 62.
3. S. Jarvis and C. Nelson-Piercy, "Management of Nausea and Vomiting in Pregnancy," *BMJ* 342 (2011): d3606.
4. A. Lacasse, E. Rey, E. Ferreira, C. Morin, and A. Bérard, "Nausea and Vomiting of Pregnancy: What About Quality of Life?," *BJOG : An International Journal of Obstetrics and Gynaecology* 115, no. 12 (2008): 1484–1493.
5. J. A. Gazmararian, R. Petersen, D. J. Jamieson, et al., "Hospitalizations During Pregnancy Among Managed Care Enrollees," *Obstetrics and Gynecology* 100, no. 1 (2002): 94–100.
6. M. Erick, J. T. Cox, and K. M. Mogensen, "American College of Obstetricians and Gynecologists Practice Bulletin 189: Nausea and Vomiting of Pregnancy," *Obstetrics and Gynecology* 131, no. 5 (2018): 935.
7. Royal College of Obstetricians and Gynaecologists, "The Management of Nausea and Vomiting of Pregnancy and Hyperemesis Gravidarum (Green-Top Guideline No. 69)," 2016.
8. A. Matthews, T. Dowswell, D. M. Haas, M. Doyle, and D. P. O'Mathúna, "Interventions for Nausea and Vomiting in Early Pregnancy," *Cochrane Database of Systematic Reviews* 2010 (2010): 9.
9. C. Hansen, T. A. Desrosiers, K. Wisniewski, M. J. Strickland, M. M. Werler, and S. M. Gilboa, "Use of Antihistamine Medications During Early Pregnancy and Selected Birth Defects: The National Birth Defects Prevention Study, 1997–2011," *Birth Defects Research* 112, no. 16 (2020): 1234–1252.
10. M. Anderka, A. A. Mitchell, C. Louik, M. M. Werler, S. Hernández-Díaz, and S. A. Rasmussen, "National Birth Defects Prevention Study.

Medications Used to Treat Nausea and Vomiting of Pregnancy and the Risk of Selected Birth Defects," *Birth Defects Research Part A: Clinical and Molecular Teratology* 94, no. 1 (2012): 22–30.

11. A. Bérard, O. Sheehy, J. Gorgui, J.-P. Zhao, C. Soares de Moura, and S. Bernatsky, "New Evidence for Concern Over the Risk of Birth Defects From Medications for Nausea and Vomiting of Pregnancy," *Journal of Clinical Epidemiology* 116 (2019): 39–48.
12. N. L. Schrager, S. E. Parker, and M. M. Werler, "The Association of Nausea and Vomiting of Pregnancy, Its Treatments, and Select Birth Defects: Findings From the National Birth Defect Prevention Study," *Birth Defects Research* 115, no. 3 (2023): 275–289.
13. Y. C. Kaplan, J. L. Richardson, E. Keskin-Arslan, H. Erol-Coskun, and D. Kennedy, "Use of Ondansetron During Pregnancy and the Risk of Major Congenital Malformations: A Systematic Review and Meta-Analysis," *Reproductive Toxicology* 86 (2019): 1–13.
14. K. F. Huybrechts, S. Hernández-Díaz, L. Straub, et al., "Association of Maternal First-Trimester Ondansetron Use With Cardiac Malformations and Oral Clefts in Offspring," *Journal of the American Medical Association* 320, no. 23 (2018): 2429–2437.
15. K. F. Huybrechts, S. Hernandez-Diaz, L. Straub, et al., "Intravenous Ondansetron in Pregnancy and Risk of Congenital Malformations," *Journal of the American Medical Association* 323, no. 4 (2020): 372–374.
16. A. Panchaud, B. Cleary, C. Weber-Schoendorfer, et al., "The Risk of Questioning the Safety of Drugs Considered Safe in Pregnancy at the Era of Big Data: The Everlasting Case of Doxylamine," *Journal of Clinical Epidemiology* 152 (2022): 125–126.
17. A. Chortatos, M. Haugen, P. O. Iversen, et al., "Pregnancy Complications and Birth Outcomes Among Women Experiencing Nausea Only or Nausea and Vomiting During Pregnancy in the Norwegian Mother and Child Cohort Study," *BMC Pregnancy and Childbirth* 15, no. 1 (2015): 138.
18. N. L. Schrager, S. E. Parker, M. M. Werler, and Study ftNBDP, "The Timing, Duration, and Severity of Nausea and Vomiting of Pregnancy and Adverse Birth Outcomes Among Controls Without Birth Defects in the National Birth Defects Prevention Study," *Birth Defects Research* 116, no. 4 (2024): e2334.
19. C. D. Chambers, D. L. Johnson, R. Xu, et al., "Birth Outcomes in Women Who Have Taken Adalimumab in Pregnancy: A Prospective Cohort Study," *PLoS One* 14, no. 10 (2019): e0223603.
20. G. Koren, R. Boskovic, M. Hard, C. Maltepe, Y. Navioz, and A. Einarson, "Motherisk-PUQE (Pregnancy-Unique Quantification of Emesis and Nausea) Scoring System for Nausea and Vomiting of Pregnancy," *American Journal of Obstetrics and Gynecology* 186, no. 5 Suppl Under-standing (2002): S228–S231.
21. Clinical Pharmacology powered by ClinicalKey, "Login Using Your Elsevier Credentials," accessed January 21, 2025, <https://www.clinicalkey.com/pharmacology/>.
22. U.S. Department of Health and Human Services—Centers for Disease Control and Prevention (CDC), "Metropolitan Atlanta Congenital Defects Program (MACDP)".
23. U.S. Department of Health and Human Services—Centers for Disease Control and Prevention (CDC), "Growth Charts".
24. A. Hollingshead, *Four Factor Index of Social Status (Unpublished Working Paper)* (Department of Sociology, Yale University, 1975).
25. G. Koren, S. Madjunkova, and C. Maltepe, "The Protective Effects of Nausea and Vomiting of Pregnancy Against Adverse Fetal Outcome—A Systematic Review," *Reproductive Toxicology* 47 (2014): 77–80.
26. S. R. Slaughter, R. Hearn-Stokes, T. Van Der Vlugt, and H. V. Joffe, "FDA Approval of Doxylamine–Pyridoxine Therapy for Use in Pregnancy," *New England Journal of Medicine* 370, no. 12 (2014): 1081–1083.

27. C. R. Dormuth, B. Winquist, A. Fisher, et al., "Comparison of Pregnancy Outcomes of Patients Treated With Ondansetron vs Alternative Antiemetic Medications in a Multinational, Population-Based Cohort," *JAMA Network Open* 4, no. 4 (2021): e215329–e215329.
28. A. Zambelli-Weiner, C. Via, M. Yuen, D. J. Weiner, and R. S. Kirby, "First Trimester Ondansetron Exposure and Risk of Structural Birth Defects," *Reproductive Toxicology* 83 (2019): 14–20.
29. E. A. Suarez, K. Boggess, S. M. Engel, T. Stürmer, J. L. Lund, and M. J. Funk, "Ondansetron Use in Early Pregnancy and the Risk of Late Pregnancy Outcomes," *Pharmacoepidemiology and Drug Safety* 30, no. 2 (2021): 114–125.
30. B. Pasternak, H. Svanström, and A. Hviid, "Ondansetron in Pregnancy and Risk of Adverse Fetal Outcomes," *New England Journal of Medicine* 368, no. 9 (2013): 814–823.
31. L. Dodds, D. B. Fell, K. S. Joseph, V. M. Allen, and B. Butler, "Outcomes of Pregnancies Complicated by Hyperemesis Gravidarum," *Obstetrics and Gynecology* 107, no. 2 Pt 1 (2006): 285–292.
32. T. J. Roseboom, A. C. J. Ravelli, J. A. van der Post, and R. C. Painter, "Maternal Characteristics Largely Explain Poor Pregnancy Outcome After Hyperemesis Gravidarum," *European Journal of Obstetrics, Gynecology, and Reproductive Biology* 156, no. 1 (2011): 56–59.
33. A. Hastoy, P. Lien Tran, O. Lakestani, G. Barau, P. Gérardin, and M. Boukerrou, "L'hyperémèse Gravidique : Quelles Conséquences Sur la Grossesse ?," *Journal de Gynécologie, Obstétrique et Biologie de la Reproduction* 44, no. 2 (2015): 154–163.
34. K. Heitmann, H. C. Svendsen, I. H. Sporsheim, and L. Holst, "Nausea in Pregnancy: Attitudes Among Pregnant Women and General Practitioners on Treatment and Pregnancy Care," *Scandinavian Journal of Primary Health Care* 34, no. 1 (2016): 13–20.
35. K. Heitmann, A. Solheimsnes, G. C. Havnen, H. Nordeng, and L. Holst, "Treatment of Nausea and Vomiting During Pregnancy—A Cross-Sectional Study Among 712 Norwegian Women," *European Journal of Clinical Pharmacology* 72, no. 5 (2016): 593–604.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** List of all the specific major congenital malformations in each live-born infant in the NVP treatment exposed ($n=36$), untreated-NVP ($n=80$), and no NVP group ($n=30$). **Table S2:** Sensitivity analysis for disease severity: comparing birth outcomes among women exposed to NVP monotherapy and polytherapy to women with untreated NVP and women without NVP. **Table S3:** Sensitivity analysis for treatment initiation at the NVP group level (A.) and for the ondansetron exposed pregnancies (B.).